

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460674.977 +/- 0.022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.147 +/- 0.034
Transit depth (Rp/Rs)^2	2.15 +/- 0.99 [%]
Orbital Inclination (inc)	84.69 +/- 2.18
Airmass coefficient 1 (a1)	1.1859 +/- 0.0073
Airmass coefficient 2 (a2)	-0.09 +/- 0.014
Scatter in the residuals of the lightcurve fit is	0.61 %
Best Comparison Star	None
Optimal Aperture	4.16
Optimal Annulus	12.35
Transit Duration (day)	0.033 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.147 $\pm$ 0.034 vs 0.1276 (deviation 0.57 σ)
Duration fit vs published	0.033 d vs 0.1179 d (deviation 3.03 σ)
Mid-time O-C	-5334.4 min
Depth SNR	4.3
Residual scatter	0.61 %

Light curve

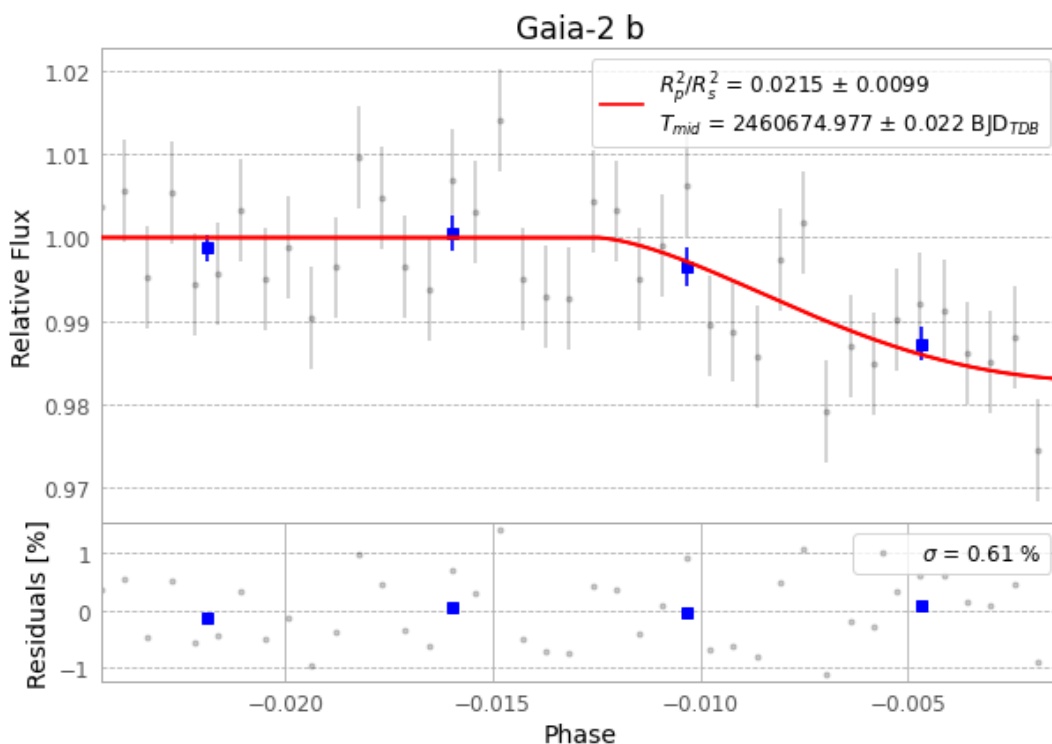


Figure 1 — FinalLightCurve_Gaia-2 b_2025-01-02.png (SHA-256 c9b3018134015043...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [414, 153], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 d97deba30792369d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

42 FITS frames (28 MB), Gaia-2250103014515.FITS → Gaia-2250103034815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Gaia-2-250103.log (06329a875cb6...), FinalLightCurve_Gaia-2 b_2025-01-02.png (c9b301813401...), FinalLightCurve_Gaia-2 b_2025-01-02.csv (046b971f06d9...)

Compute resources

Wall time 77.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-13 22:48Z.